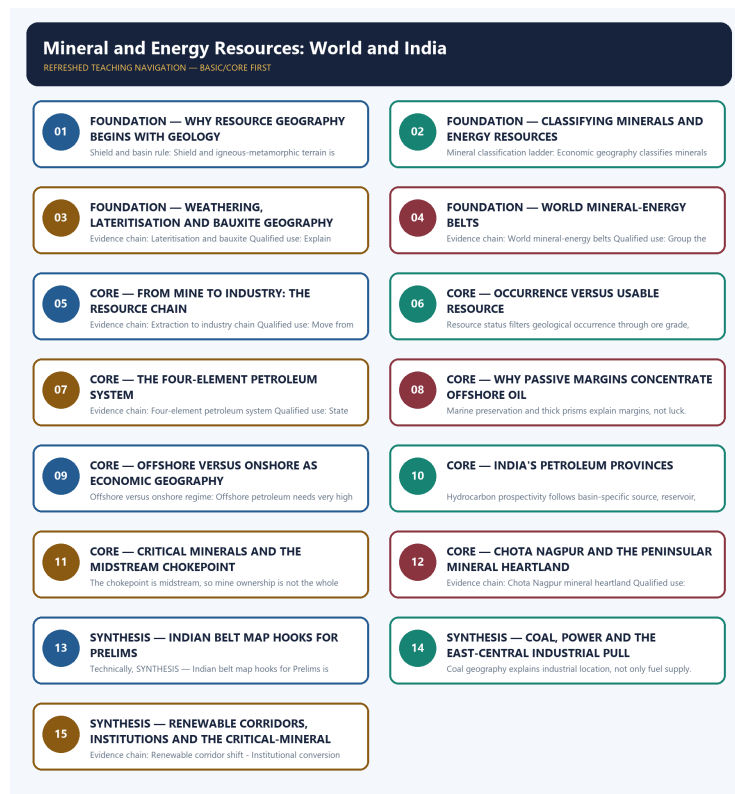


SOLVED PRACTICE WORKBOOK

Mineral and Energy Resources: World and India - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly explains Geological control of distribution?

A. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. B. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. C. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. D. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels.

Answer: A. Explanation: Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. The other options describe different processes, locations, scales or governance categories.

Q2. Which option is the safest spatial interpretation of Geological control of distribution?

A. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. B. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. C. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. D. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow.

Answer: B. Explanation: Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. The other options describe different processes, locations, scales or governance categories.

Q3. Which statement preserves the process boundary for Geological control of distribution?

A. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. B. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. C. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. D. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas.

Answer: C. Explanation: Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. The other options describe different processes, locations, scales or governance categories.

Q4. Which option avoids the main UPSC trap concerning Geological control of distribution?

A. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. B. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. C. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. D. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes.

Answer: D. Explanation: Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. The other options describe different processes, locations, scales or governance categories.

Q5. Which statement correctly explains Mineral classification ladder?

A. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. B. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. C. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. D. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map.

Answer: A. Explanation: Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. The other options describe different processes, locations, scales or governance categories.

Q6. Which option is the safest spatial interpretation of Mineral classification ladder?

A. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. B. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. C. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. D. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones.

Answer: B. Explanation: Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. The other options describe different processes, locations, scales or governance categories.

Q7. Which statement preserves the process boundary for Mineral classification ladder?

A. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. B. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. C. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. D. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map.

Answer: C. Explanation: Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. The other options describe different processes, locations, scales or governance categories.

Q8. Which option avoids the main UPSC trap concerning Mineral classification ladder?

A. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. B. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. C. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. D. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels.

Answer: D. Explanation: Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels.

The other options describe different processes, locations, scales or governance categories.

Q9. Which statement correctly explains Energy classification pairs?

A. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. B. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. C. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. D. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas.

Answer: A. Explanation: Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. The other options describe different processes, locations, scales or governance categories.

Q10. Which option is the safest spatial interpretation of Energy classification pairs?

A. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. B. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. C. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. D. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map.

Answer: B. Explanation: Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. The other options describe different processes, locations, scales or governance categories.

Q11. Which statement preserves the process boundary for Energy classification pairs?

A. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. B. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. C. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. D. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.

Answer: C. Explanation: Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. The other options describe different processes, locations, scales or governance categories.

Q12. Which option avoids the main UPSC trap concerning Energy classification pairs?

A. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. B. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. C. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. D. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow.

Answer: D. Explanation: Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. The other options describe different processes, locations, scales or governance categories.

Q13. Which statement correctly explains Shield and basin rule?

A. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. B. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. C. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. D. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map.

Answer: A. Explanation: Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. The other options describe different processes, locations, scales or governance categories.

Q14. Which option is the safest spatial interpretation of Shield and basin rule?

A. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. B. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. C. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. D. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.

Answer: B. Explanation: Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. The other options describe different processes, locations, scales or governance categories.

Q15. Which statement preserves the process boundary for Shield and basin rule?

A. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. B. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. C. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. D. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region.

Answer: C. Explanation: Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. The other options describe different processes, locations, scales or governance categories.

Q16. Which option avoids the main UPSC trap concerning Shield and basin rule?

A. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. B. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. C. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. D. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas.

Answer: D. Explanation: Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. The other options describe different processes, locations, scales or governance categories.

Q17. Which statement correctly explains Lateritisation and bauxite?

A. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. B. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. C. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. D. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.

Answer: A. Explanation: Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. The other options describe different processes, locations, scales or governance categories.

Q18. Which option is the safest spatial interpretation of Lateritisation and bauxite?

A. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. B. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. C. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. D. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.

Answer: B. Explanation: Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. The other options describe different processes, locations, scales or governance categories.

Q19. Which statement preserves the process boundary for Lateritisation and bauxite?

A. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. B. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. C. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. D. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.

Answer: C. Explanation: Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. The other options describe different processes, locations, scales or governance categories.

Q20. Which option avoids the main UPSC trap concerning Lateritisation and bauxite?

A. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. B. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. C. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. D. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map.

Answer: D. Explanation: Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. The other options describe different processes, locations, scales or governance categories.

Q21. Which statement correctly explains World mineral-energy belts?

A. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. B. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. C. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. D. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region.

Answer: A. Explanation: The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. The other options describe different processes, locations, scales or governance categories.

Q22. Which option is the safest spatial interpretation of World mineral-energy belts?

A. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. B. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. C. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. D. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum.

Answer: B. Explanation: The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. The other options describe different processes, locations, scales or governance categories.

Q23. Which statement preserves the process boundary for World mineral-energy belts?

A. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. B. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. C. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. D. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum.

Answer: C. Explanation: The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. The other options describe different processes, locations, scales or governance categories.

Q24. Which option avoids the main UPSC trap concerning World mineral-energy belts?

A. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. B. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. C. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. D. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones.

Answer: D. Explanation: The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. The other options describe different processes, locations, scales or governance categories.

Q25. Which statement correctly explains Extraction to industry chain?

A. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. B. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. C. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. D. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps.

Answer: A. Explanation: Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. The other options describe different processes, locations, scales or governance categories.

Q26. Which option is the safest spatial interpretation of Extraction to industry chain?

A. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. B. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. C. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. D. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps.

Answer: B. Explanation: Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. The other options describe different processes, locations, scales or governance categories.

Q27. Which statement preserves the process boundary for Extraction to industry chain?

A. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. B. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. C. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. D. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not.

Answer: C. Explanation: Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. The other options describe different processes, locations, scales or governance categories.

Q28. Which option avoids the main UPSC trap concerning Extraction to industry chain?

A. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. B. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. C. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. D. Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region.

Answer: D. Explanation: Minerals matter along a chain of extraction, transport, processing, power supply and industrial clustering, so a mine site alone does not create an industrial region. The other options describe different processes, locations, scales or governance categories.

Q29. Which statement correctly explains Occurrence versus usable resource?

A. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. B. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. C. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. D. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not.

Answer: A. Explanation: Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. The other options describe different processes, locations, scales or governance categories.

Q30. Which option is the safest spatial interpretation of Occurrence versus usable resource?

A. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. B. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. C. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. D. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps.

Answer: B. Explanation: Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. The other options describe different processes, locations, scales or governance categories.

Q31. Which statement preserves the process boundary for Occurrence versus usable resource?

A. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. B. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. C. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. D. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine.

Answer: C. Explanation: Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. The other options describe different processes, locations, scales or governance categories.

Q32. Which option avoids the main UPSC trap concerning Occurrence versus usable resource?

A. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. B. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. C. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. D. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.

Answer: D. Explanation: Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all. The other options describe different processes, locations, scales or governance categories.

Q33. Which statement correctly explains Four-element petroleum system?

A. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. B. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. C. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. D. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins.

Answer: A. Explanation: Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. The other options describe different processes, locations, scales or governance categories.

Q34. Which option is the safest spatial interpretation of Four-element petroleum system?

A. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. B. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. C. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. D. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine.

Answer: B. Explanation: Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. The other options describe different processes, locations, scales or governance categories.

Q35. Which statement preserves the process boundary for Four-element petroleum system?

A. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. B. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. C. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. D. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins.

Answer: C. Explanation: Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. The other options describe different processes, locations, scales or governance categories.

Q36. Which option avoids the main UPSC trap concerning Four-element petroleum system?

A. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. B. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. C. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. D. Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum.

Answer: D. Explanation: Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum. The other options describe different processes, locations, scales or governance categories.

Q37. Which statement correctly explains Why margins concentrate oil?

A. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. B. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. C. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. D. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins.

Answer: A. Explanation: Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. The other options describe different processes, locations, scales or governance categories.

Q38. Which option is the safest spatial interpretation of Why margins concentrate oil?

A. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. B. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. C. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. D. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies.

Answer: B. Explanation: Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. The other options describe different processes, locations, scales or governance categories.

Q39. Which statement preserves the process boundary for Why margins concentrate oil?

A. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. B. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. C. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. D. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine.

Answer: C. Explanation: Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. The other options describe different processes, locations, scales or governance categories.

Q40. Which option avoids the main UPSC trap concerning Why margins concentrate oil?

A. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. B. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. C. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. D. Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps.

Answer: D. Explanation: Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps. The other options describe different processes, locations, scales or governance categories.

Q41. Which statement correctly explains Offshore versus onshore regime?

A. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. B. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. C. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. D. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins.

Answer: A. Explanation: Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. The other options describe different processes, locations, scales or governance categories.

Q42. Which option is the safest spatial interpretation of Offshore versus onshore regime?

A. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. B. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. C. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. D. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine.

Answer: B. Explanation: Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. The other options describe different processes, locations, scales or governance categories.

Q43. Which statement preserves the process boundary for Offshore versus onshore regime?

A. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. B. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. C. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. D. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies.

Answer: C. Explanation: Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. The other options describe different processes, locations, scales or governance categories.

Q44. Which option avoids the main UPSC trap concerning Offshore versus onshore regime?

A. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. B. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. C. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. D. Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not.

Answer: D. Explanation: Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not. The other options describe different processes, locations, scales or governance categories.

Q45. Which statement correctly explains Indian petroleum provinces?

A. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. B. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. C. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. D. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base.

Answer: A. Explanation: India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. The other options describe different processes, locations, scales or governance categories.

Q46. Which option is the safest spatial interpretation of Indian petroleum provinces?

A. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. B. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. C. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. D. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha.

Answer: B. Explanation: India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. The other options describe different processes, locations, scales or governance categories.

Q47. Which statement preserves the process boundary for Indian petroleum provinces?

A. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. B. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. C. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. D. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.

Answer: C. Explanation: India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. The other options describe different processes, locations, scales or governance categories.

Q48. Which option avoids the main UPSC trap concerning Indian petroleum provinces?

A. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. B. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. C. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. D. India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins.

Answer: D. Explanation: India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins. The other options describe different processes, locations, scales or governance categories.

Q49. Which statement correctly explains Midstream critical-mineral chokepoint?

A. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. B. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. C. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. D. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies.

Answer: A. Explanation: For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. The other options describe different processes, locations, scales or governance categories.

Q50. Which option is the safest spatial interpretation of Midstream critical-mineral chokepoint?

A. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. B. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. C. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. D. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha.

Answer: B. Explanation: For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. The other options describe different processes, locations, scales or governance categories.

Q51. Which statement preserves the process boundary for Midstream critical-mineral chokepoint?

A. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. B. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. C. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. D. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.

Answer: C. Explanation: For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is

midstream rather than at the mine. The other options describe different processes, locations, scales or governance categories.

Q52. Which option avoids the main UPSC trap concerning Midstream critical-mineral chokepoint?

A. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. B. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. C. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. D. For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine.

Answer: D. Explanation: For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine. The other options describe different processes, locations, scales or governance categories.

Q53. Which statement correctly explains Critical-mineral response levers?

A. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. B. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. C. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. D. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.

Answer: A. Explanation: Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. The other options describe different processes, locations, scales or governance categories.

Q54. Which option is the safest spatial interpretation of Critical-mineral response levers?

A. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. B. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. C. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. D. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.

Answer: B. Explanation: Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. The other options describe different processes, locations, scales or governance categories.

Q55. Which statement preserves the process boundary for Critical-mineral response levers?

A. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. B. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. C. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. D. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography.

Answer: C. Explanation: Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. The other options describe

different processes, locations, scales or governance categories.

Q56. Which option avoids the main UPSC trap concerning Critical-mineral response levers?

A. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. B. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. C. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. D. Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies.

Answer: D. Explanation: Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies. The other options describe different processes, locations, scales or governance categories.

Q57. Which statement correctly explains Chota Nagpur mineral heartland?

A. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. B. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. C. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. D. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.

Answer: A. Explanation: Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. The other options describe different processes, locations, scales or governance categories.

Q58. Which option is the safest spatial interpretation of Chota Nagpur mineral heartland?

A. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. B. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. C. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. D. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map.

Answer: B. Explanation: Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. The other options describe different processes, locations, scales or governance categories.

Q59. Which statement preserves the process boundary for Chota Nagpur mineral heartland?

A. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. B. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. C. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. D. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling.

Answer: C. Explanation: Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. The other options describe different processes, locations, scales or governance categories.

Q60. Which option avoids the main UPSC trap concerning Chota Nagpur mineral heartland?

A. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. B. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. C. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. D. Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base.

Answer: D. Explanation: Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base. The other options describe different processes, locations, scales or governance categories.

Q61. Which statement correctly explains Indian belt map hooks?

A. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. B. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. C. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. D. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.

Answer: A. Explanation: Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. The other options describe different processes, locations, scales or governance categories.

Q62. Which option is the safest spatial interpretation of Indian belt map hooks?

A. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. B. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. C. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. D. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography.

Answer: B. Explanation: Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. The other options describe different processes, locations, scales or governance categories.

Q63. Which statement preserves the process boundary for Indian belt map hooks?

A. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. B. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. C. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. D. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling.

Answer: C. Explanation: Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. The other options describe different processes, locations, scales or governance categories.

Q64. Which option avoids the main UPSC trap concerning Indian belt map hooks?

A. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. B. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. C. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. D. Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha.

Answer: D. Explanation: Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Agucha. The other options describe different processes, locations, scales or governance categories.

Q65. Which statement correctly explains Coal and the east-central pull?

A. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. B. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. C. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. D. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography.

Answer: A. Explanation: Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. The other options describe different processes, locations, scales or governance categories.

Q66. Which option is the safest spatial interpretation of Coal and the east-central pull?

A. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. B. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. C. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. D. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling.

Answer: B. Explanation: Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. The other options describe different processes, locations, scales or governance categories.

Q67. Which statement preserves the process boundary for Coal and the east-central pull?

A. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. B. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. C. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. D. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels.

Answer: C. Explanation: Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. The other options describe different processes, locations, scales or governance categories.

Q68. Which option avoids the main UPSC trap concerning Coal and the east-central pull?

A. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. B. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. C. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. D. Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.

Answer: D. Explanation: Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India. The other options describe different processes, locations, scales or governance categories.

Q69. Which statement correctly explains Renewable corridor shift?

A. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. B. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. C. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. D. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes.

Answer: A. Explanation: Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. The other options describe different processes, locations, scales or governance categories.

Q70. Which option is the safest spatial interpretation of Renewable corridor shift?

A. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. B. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. C. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. D. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling.

Answer: B. Explanation: Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the

renewable map does not mirror the coal map. The other options describe different processes, locations, scales or governance categories.

Q71. Which statement preserves the process boundary for Renewable corridor shift?

A. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. B. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. C. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. D. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes.

Answer: C. Explanation: Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. The other options describe different processes, locations, scales or governance categories.

Q72. Which option avoids the main UPSC trap concerning Renewable corridor shift?

A. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. B. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. C. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. D. Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map.

Answer: D. Explanation: Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map. The other options describe different processes, locations, scales or governance categories.

Q73. Which statement correctly explains Institutional conversion ladder?

A. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. B. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. C. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. D. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling.

Answer: A. Explanation: The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. The other options describe different processes, locations, scales or governance categories.

Q74. Which option is the safest spatial interpretation of Institutional conversion ladder?

A. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. B. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. C. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. D. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels.

Answer: B. Explanation: The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. The other options describe different processes, locations, scales or governance categories.

Q75. Which statement preserves the process boundary for Institutional conversion ladder?

A. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. B. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. C. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. D. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow.

Answer: C. Explanation: The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. The other options describe different processes, locations, scales or governance categories.

Q76. Which option avoids the main UPSC trap concerning Institutional conversion ladder?

A. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. B. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. C. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. D. The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography.

Answer: D. Explanation: The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography. The other options describe different processes, locations, scales or governance categories.

Q77. Which statement correctly explains National Critical Mineral Mission anchor?

A. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. B. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels. C. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. D. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow.

Answer: A. Explanation: The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. The other options describe different processes, locations, scales or governance categories.

Q78. Which option is the safest spatial interpretation of National Critical Mineral Mission anchor?

A. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow. B. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. C. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. D. Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels.

Answer: B. Explanation: The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. The other options describe different processes, locations, scales or governance categories.

Q79. Which statement preserves the process boundary for National Critical Mineral Mission anchor?

A. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. B. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. C. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. D. Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow.

Answer: C. Explanation: The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. The other options describe different processes, locations, scales or governance categories.

Q80. Which option avoids the main UPSC trap concerning National Critical Mineral Mission anchor?

A. The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones. B. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. C. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. D. The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling.

Answer: D. Explanation: The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling. The other options describe different processes, locations, scales or governance categories.

PYQS AND ANSWER PRACTICE

VERIFIED PYQ OWNERSHIP AUDIT

Geography Topic 31 owns direct Mains PYQ demand in the audited routing ledgers. Five GS-I demands are routed to this owner: 2018 Q5, 2021 Q5, 2021 Q16, 2022 Q6 and 2025 Q14. Each is answered below as an original model solution built only from the owner evidence. The routed Prelims demands for this topic are recorded in the owner ledgers as objective questions whose official keys are either unavailable locally or deliberately not inferred, so no option letter, answer key or invented question wording is reproduced here.

OWNER PYQ LEDGER EXTRACTS

9. PYQ / exam linkage

- ANALYSIS: Recurring Prelims theme: classify resources as **ferrous / non-ferrous / non-metallic / energy**.
- ANALYSIS: Recurring Prelims theme: pair a resource with the right **geological setting**.
- ANALYSIS: Recurring GS-I/GS-III theme: explain how resource distribution influences **industrial regions, trade routes and strategic vulnerability**.
- FACT: **2025 GS-I direct PYQs:** (i) ecological/economic benefits of solar generation in India; (ii) geographic distribution of **offshore oil** and how it differs from onshore occurrence.
- ANALYSIS: Offshore petroleum clusters on submerged continental-margin sedimentary basins with source/reservoir/seal structures; onshore reserves occur in terrestrial sedimentary basins. Exact route: . . /README . md.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2024-2025 . md, _PYQ-ROUTING-PRELIMS-2024-2025 . md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 4

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	Prelims GS-I	88	Sumed pipeline (Persian Gulf oil to Europe; Red Sea-Mediterranean)	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2024	Prelims GS-I	97	Venezuela and the world's largest proven oil reserves	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	GS-I	14	Distribution of off-shore oil reserves versus on-shore occurrences	Explain · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2025	Prelims GS-I	79	Resource-rich country pairs (Botswana-diamond, Chile-lithium, Indonesia-nickel)	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Sumed pipeline (Persian Gulf oil to Europe; Red Sea-Mediterranean)
- Venezuela and the world's largest proven oil reserves

- Distribution of off-shore oil reserves versus on-shore occurrences
- Resource-rich country pairs (Botswana-diamond, Chile-lithium, Indonesia-nickel)

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-PRELIMS-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2023
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 7

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	Prelims GS-I	27	State government mineral mining rights and distribution India	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2019	Prelims GS-I	62	Minor minerals management and state government powers	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2020	Prelims GS-I	53	West Texas Intermediate crude oil grade in news	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2020	Prelims GS-I	92	Major minerals designation Bentonite Chromite Kyanite Sillimanite India	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	Prelims GS-I	6	Ilmenite and rutile minerals and titanium source	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	Prelims GS-I	7	Global cobalt production leading country for EV batteries	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	Prelims GS-I	66	Coal-based thermal power plants India seawater water-stress	Objective question; official key unavailable locally	Cross-routed to spatial-resource and thermal-infrastructure owners; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- State government mineral mining rights and distribution India
- Minor minerals management and state government powers
- West Texas Intermediate crude oil grade in news
- Major minerals designation Bentonite Chromite Kyanite Sillimanite India
- Ilmenite and rutile minerals and titanium source
- Global cobalt production leading country for EV batteries
- Coal-based thermal power plants India seawater water-stress

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

7. PYQ / exam linkage

- ANALYSIS: Recurring Prelims theme: locate **Chota Nagpur**, Aravallis, offshore basins and coastal placer belts.
- ANALYSIS: Recurring Prelims theme: distinguish **metallic ore belts** from **hydrocarbon basins**.
- ANALYSIS: Recurring GS-III theme: analyse India's **critical-mineral vulnerability** in the clean-energy transition.
- ANALYSIS: Recurring GS-I/GS-III theme: relate mineral distribution to steel plants, power plants and industrial corridors.
- FACT: **2025 GS-I direct PYQs**: solar generation's ecological/economic

benefits, and the geographic explanation of offshore oil distribution versus onshore occurrence.

- FACT: **2025 GS-III cross-route**: why mining is an environmental hazard and

what remedial measures are needed. Keep mineral occurrence/extraction geography here; route EIA, reclamation and mine-closure governance to

.../Environment-and-Ecology/advanced/16_Environmental-Impact-Assessment-and-NGT.md.

- ANALYSIS: Build the oil answer from sedimentary-basin evolution, continental

shelves/margins, source-reservoir-seal geometry and exploration/access differences-not from a memorised field list.

Exact route: .../README.md.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance**: Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md.

- **Years represented**: 2018, 2021, 2022
- **Paper(s)**: GS-I
- **Routed question demands**: 4

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	GS-I	5	India's interest in the resources of the Arctic region	Why · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2021	GS-I	5	Gondwanaland mineral base and mining's low share in GDP	Discuss · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2021	GS-I	16	Uneven distribution of mineral oil in the world	Discuss · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	GS-I	6	Natural resource potentials of the Deccan Trap	Discuss · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- India's interest in the resources of the Arctic region
- Gondwanaland mineral base and mining's low share in GDP
- Uneven distribution of mineral oil in the world
- Natural resource potentials of the Deccan Trap

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2018 GS-I Q5 (Why, 10 marks, 150 words)

Demand: Explain why India has an interest in the resources of the Arctic region.

Status: Verified routed demand from the audited Mains routing ledger for 2018-2023; the owner file records it as routed to this topic. The wording is the ledger's neutral rendering, not a reproduced official paper text.

Model solution: Frame the Arctic as a high-latitude resource frontier whose hydrocarbon potential rests on sedimentary basins and continental-margin geology, exactly the shield-versus-basin logic that governs resource distribution everywhere. India's interest follows the same chain that this owner sets out: occurrence must be converted into usable resource through transport, processing and power, so the Arctic matters for prospective fuel and mineral access, for shipping routes that shorten the distance filter, and for scientific presence that supports exploration capability. Qualify the answer twice: occurrence is not usability, since grade, depth, distance, technology and price still decide viability; and no reserve, share or output figure should be quoted, because the owner explicitly forbids remembered figures. Conclude that the interest is strategic and anticipatory rather than an established production relationship.

PYQ DEMAND CARD 2 - 2021 GS-I Q5 (Discuss, 10 marks, 150 words)

Demand: Discuss why India's Gondwanaland mineral base has not translated into a large share of mining in the national economy.

Status: Verified routed demand from the audited Mains routing ledger for 2018-2023; the owner file records it as routed to this topic. Only the ledger's neutral rendering is used.

Model solution: Open with the endowment: Gondwana sequences in the Damodar, Mahanadi, Son and Godavari valley basins, set within the peninsular shield that Khullar treats as India's mineral heartland, give India a genuinely strong bulk-mineral base. Then apply this owner's central distinction between occurrence and usable resource. Endowment alone is inert: grade, depth, transport distance, technology and price stand between a deposit and an economically significant industry, and the chain from extraction through transport, processing and power to industrial clustering has to be completed before value is captured. East-central abundance therefore raises a logistics question of rail evacuation, slurry pipelines, port linkage and power availability rather than an automatic economic dividend, and bulk-mineral strength does not by itself create high-end processing strength. Qualify by refusing any GDP-share, production or reserve figure from memory, and conclude that the low economic share reflects an unconverted chain rather than a poor endowment.

PYQ DEMAND CARD 3 - 2021 GS-I Q16 (Discuss, 15 marks, 250 words)

Demand: Discuss the uneven distribution of mineral oil in the world.

Status: Verified routed demand from the audited Mains routing ledger for 2018-2023; the owner file records it as routed to this topic. No official answer key exists for a Mains question, and none is implied.

Model solution: Thesis: mineral oil is unevenly distributed because it requires a rare coincidence of geological conditions, not because it is randomly scattered. Establish the four-element petroleum system first, since source rock, reservoir rock, seal and trap must occur together, which makes a sedimentary basin necessary but never sufficient. Explain next why particular settings satisfy all four conditions: continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, passive margins subside steadily and accumulate thick undisturbed prisms, deltas deliver source and reservoir facies together, and salt tectonics generates both traps and seals. Then locate the pattern: the Persian Gulf as the classic sedimentary hydrocarbon province, the Russian and Siberian basins as a continental-scale frontier, the interior United States belts, and India's own combination of Assam-Arakan and Cambay onshore basins with a western offshore province and Krishna-Godavari and Cauvery deltaic gas. Add the second filter: uplift, erosion and faulting have breached many onshore accumulations, so preservation as well as formation shapes the map. Qualify by separating occurrence from usable resource and by refusing reserve or production percentages from memory. Conclude that the world oil map is a preserved record of basin evolution read through an economic filter.

PYQ DEMAND CARD 4 - 2022 GS-I Q6 (Discuss, 10 marks, 150 words)

Demand: Discuss the natural resource potentials of the Deccan Trap region.

Status: Verified routed demand from the audited Mains routing ledger for 2018-2023; the owner file records it as routed to this topic. The rendering is the ledger's neutral phrasing.

Model solution: Treat the Deccan Trap as a case of this owner's central rule that resource potential is deduced from geological and weathering history. As a vast basaltic province it does not carry shield-core metallic ore belts or thick hydrocarbon-bearing sedimentary sequences; its potential is instead of the kinds that basalt and its weathering profile generate. Tropical weathering and lateritisation on basaltic terrain concentrate bauxite on suitable plateaux and hilltops, and the same weathering produces deep regur soils that convert the region into an agricultural rather than a metallurgical resource base. Basalt also yields large-volume building and industrial stone, while jointed and fractured flows govern the groundwater question. Qualify sharply: do not attribute coal, petroleum or shield-type metallic ore belts to the traps, and do not attach reserve or production figures from memory. Conclude that the Deccan Trap's potential is weathering-derived and surface-linked, which is precisely why it must not be read through the Chota Nagpur template.

PYQ DEMAND CARD 5 - 2025 GS-I Q14 (Explain, 15 marks, 250 words)

Demand: Explain the distribution of offshore oil reserves and how it differs from onshore occurrence.

Status: Verified routed demand from the audited Mains routing ledger for 2024-2025, routed to this owner as the owning topic. The owner file itself flags the demand and supplies the geological and economic logic.

Model solution: Thesis: offshore petroleum is not onshore petroleum placed under water; it reflects a different depositional and tectonic setting and imposes a different economic, environmental and jurisdictional regime. Begin with the common requirement, the four-element petroleum system of source, reservoir, seal and trap, which applies identically in both settings. Then explain why submerged margins are favoured: shelves and slopes are the world's great sediment sinks, marine bottom water preserves organic matter, passive margins subside quietly and build thick prisms, deltas pair source with reservoir under rapid burial and growth faulting, salt tectonics supplies traps and seals, and offshore sections escape the uplift and erosion that have breached many onshore accumulations. Next compare by axis rather than by location: geological setting, exploration method, capital and technology intensity, the field-size threshold that governs viability, infrastructure from platforms and subsea lines to coastal landfall terminals, land acquisition against fisheries and shipping conflict, contamination against marine spill risk, hazard exposure, jurisdictional regime and decommissioning obligation. Illustrate with India's own expression: Assam-Arakan and Cambay onshore against a carbonate-dominated western offshore province and deltaic Krishna-Godavari and Cauvery gas, with landfall pulling refining and petrochemicals to the coast. Qualify with the maritime-zone dimension and refuse remembered figures. Conclude with a graded verdict: offshore raises capital and marine environmental stakes while reducing land and displacement burden, so the two are a trade-off in kind rather than a ranking.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain why the world distribution of mineral and energy resources is better treated as a deduction from Earth history than as a list of producing countries. Answer in about 150 words.

Model thesis: Distribution follows geological history because shields concentrate metallic ores, sedimentary basins hold fuels and lateritic weathering profiles concentrate bauxite, while accessibility and economics decide which occurrences become usable resources.

Claim -> named evidence -> analysis -> qualification:

- Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes.
- Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas.
- Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map.
- Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.

Qualified conclusion: Distribution follows geological history because shields concentrate metallic ores, sedimentary basins hold fuels and lateritic weathering profiles concentrate bauxite, while accessibility and economics decide which occurrences become usable resources.

ORIGINAL MAINS 2 - 10 MARKS

Question: Distinguish mineral-resource geography from energy geography with suitable Indian examples. Answer in about 150 words.

Model thesis: Mineral geography asks where ores and industrial minerals occur, while energy geography asks where usable power can be produced, so India's shield-based ore belts and its solar and wind corridors do not overlap.

Claim -> named evidence -> analysis -> qualification:

- Economic geography classifies minerals as ferrous metallic, non-ferrous metallic, non-metallic industrial and mineral fuels, so coal and petroleum are mineral resources and not merely fuels.
- Energy resources are classified twice over: conventional commercial against non-conventional newer sources, and exhaustible stock against renewable flow.
- Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map.

Qualified conclusion: Mineral geography asks where ores and industrial minerals occur, while energy geography asks where usable power can be produced, so India's shield-based ore belts and its solar and wind corridors do not overlap.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain the distribution of offshore petroleum reserves and how it differs from onshore occurrence. Answer in about 250 words.

Model thesis: Offshore petroleum reflects a distinct depositional and tectonic setting on thick passive-margin sediment prisms, and it imposes a different capital, technological, environmental and jurisdictional regime rather than being onshore petroleum placed under water.

Claim -> named evidence -> analysis -> qualification:

- Oil and gas require source rock, reservoir rock, an impermeable seal and a trap geometry together, so a sedimentary basin is necessary but not sufficient for petroleum.
- Continental shelves and slopes act as great sediment sinks, oxygen-poor marine bottom water preserves organic matter, and passive margins accumulate thick undisturbed sediment prisms with deltaic and salt-tectonic traps.
- Offshore petroleum needs very high lumpy capital and large fields, works through platforms, subsea pipelines and coastal landfall terminals, and carries marine spill risk and maritime jurisdictional questions that onshore fields do not.
- India combines onshore Assam-Arakan and Cambay basins with a major western offshore province and east-coast deltaic offshore gas in the Krishna-Godavari and Cauvery margins.

Qualified conclusion: Offshore petroleum reflects a distinct depositional and tectonic setting on thick passive-margin sediment prisms, and it imposes a different capital, technological, environmental and jurisdictional regime rather than being onshore petroleum placed under water.

ORIGINAL MAINS 4 - 15 MARKS

Question: Examine why India's mineral-industrial geography still carries the imprint of the peninsular shield. Answer in about 250 words.

Model thesis: The peninsular shield and its Gondwana basins supplied the ore and coal combinations around which steel, thermal power and rail-linked industry clustered, so the historic mineral heartland continues to shape industrial location.

Claim -> named evidence -> analysis -> qualification:

- The standard world belts are the Persian Gulf hydrocarbon province, the Appalachian and interior United States belts, the Ruhr-Lorraine coal-iron core, the Russian and Siberian basins, the Chinese northern and interior coal belts, the African Copperbelt and South African shield, the Australian shield and basins, and the Andean and Latin American shield zones.
- Khullar treats the Chota Nagpur Plateau complex and the wider peninsular shield as India's classic mineral heartland and heavy-industry base.
- Iron ore belongs to Odisha-Jharkhand, Bailadila and Ballari-Hospet; Gondwana coal to the Damodar, Mahanadi, Son and Godavari valley basins; copper to Khetri, Singhbhum and Malanjkhand; and lead-zinc to Zawar and Rampura-Aguicha.
- Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.

Qualified conclusion: The peninsular shield and its Gondwana basins supplied the ore and coal combinations around which steel, thermal power and rail-linked industry clustered, so the historic mineral heartland continues to shape industrial location.

ORIGINAL MAINS 5 - 20 MARKS

Question: In the energy transition, is the critical-mineral constraint located at the mine or in the midstream? Analyse. Answer in about 300 words.

Model thesis: Extraction is geologically concentrated, but processing and refining are more concentrated still and often located elsewhere, so supply security depends on midstream capacity, recycling and substitution rather than on mine ownership alone.

Claim -> named evidence -> analysis -> qualification:

- Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.
- For transition minerals the processing and refining stage is even more geographically concentrated than extraction and often sits in different countries, so the binding chokepoint is midstream rather than at the mine.
- Recycling, substitution, supply diversification and strategic stockpiling are the principal long-term levers available to resource-poor consuming economies.
- The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling.

Qualified conclusion: Extraction is geologically concentrated, but processing and refining are more concentrated still and often located elsewhere, so supply security depends on midstream capacity, recycling and substitution rather than on mine ownership alone.

ORIGINAL MAINS 6 - 20 MARKS

Question: Assess how far renewable-energy corridors are overlaying, rather than replacing, India's coal-centred energy geography. Answer in about 300 words.

Model thesis: Renewable corridors are creating a new western and southern energy map while coal-linked east-central geography still anchors dispatchable power and freight-dependent heavy industry, so the correct verdict is overlay with partial substitution.

Claim -> named evidence -> analysis -> qualification:

- Coalfields of the Damodar valley and the adjoining eastern belt pulled steel plants, thermal power and rail-linked heavy industry into east-central India.
- Solar capacity concentrates in the dry high-insolation belts of Rajasthan and Gujarat while wind concentrates along Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes, so the renewable map does not mirror the coal map.
- The Ministry of Mines, Geological Survey of India, Indian Bureau of Mines, Ministry of Coal, Ministry of Power with the Central Electricity Authority, and NPCIL convert geological occurrence into extractive, power and siting geography.
- The Union Cabinet approved the National Critical Mineral Mission on 29 January 2025 with a total envisaged outlay of Rs 34,300 crore over seven years including Rs 16,300 crore of government expenditure, covering exploration, mining, beneficiation, processing, recovery and recycling.

Qualified conclusion: Renewable corridors are creating a new western and southern energy map while coal-linked east-central geography still anchors dispatchable power and freight-dependent heavy industry, so the correct verdict is overlay with partial substitution.